29/01/2025
Morning

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Medical | IIT-JEE | Foundations

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Answers & Solutions

Time : 3 hrs. *for* M.M. : 300

JEE (Main)-2025 Phase-1
[Computer Based Test (CBT) mode]

(Mathematics, Physics and Chemistry)

IMPORTANT INSTRUCTIONS:

(1) The test is of **3 hours** duration.

(2) This test paper consists of 75 questions. Each subject (MPC) has 25 questions. The maximum marks are 300.

(3) This question paper contains **Three** Parts. **Part-A** is Physics, **Part-B** is Chemistry and **Part-C** is **Mathematics**. Each part has only two sections: **Section-A** and **Section-B**.

(4) **Section - A :** Attempt all questions.

(5) **Section - B :** Attempt all questions.

(6) **Section - A (01 – 20)** contains 20 multiple choice questions which have **only one correct answer**. Each question carries **+4 marks** for correct answer and **–1 mark** for wrong answer.

(7) **Section - B (21 – 25) contains** 5 **Numerical value** based questions. The answer to each question should be rounded off to the **nearest integer**. Each question carries **+4 marks** for correct answer and **–1 mark** for wrong answer.



MATHEMATICS

SECTION - A

Multiple Choice Questions: This section contains 20 multiple choice questions. Each question has 4 choices (1), (2), (3) and (4), out of which **ONLY ONE** is correct.

Choose the correct answer:

1. Define a relation R on the interval $\left[0, \frac{\pi}{2}\right)$ by $x\ R\ y$ if and only if $\sec^2 x - \tan^2 y = 1$. Then R is:

(1) both reflexive and transitive but not symmetric

(2) an equivalence relation

(3) reflexive but neither symmetric not transitive

(4) both reflexive and symmetric but not transitive

Answer (2)

Sol. $x\ R\ y : \sec^2 x - \tan^2 y = 1$

Check reflexive:

$x\ R\ x = \sec^2 x - \tan^2 x = 1$

$\forall x \in \left[0, \frac{\pi}{2}\right)$

Check symmetric

$x\ R\ y \Rightarrow y\ R\ x$

$\sec^2 x - \tan^2 y = 1$

$= \sec^2 y - \tan^2 x = (1 + \tan^2 y) - (\sec^2 x - 1)$

$= 2 - (\sec^2 x - \tan^2 y)$

$= 2 - 1 = 1$

$\Rightarrow y\ R\ x$

Check transitive

$x\ R\ y$ and $y\ R\ z$

$\Rightarrow \sec^2 x - \tan^2 y = 1$

$\sec^2 y - \tan^2 z = 1$

Add $\Rightarrow \sec^2 x - \tan^2 z + (\sec^2 y - \tan^2 y) = 2$

$\Rightarrow \sec^2 x - \tan^2 z + 1 = 2 \Rightarrow x\ R\ z$

$\Rightarrow$ R is an equivalence relation.

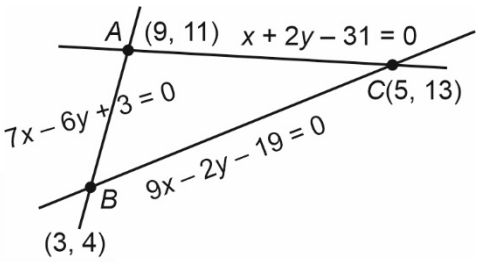
2. Let ABC be a triangle formed by the lines $7x - 6y + 3 = 0$, $x + 2y - 31 = 0$ and $9x - 2y - 19 = 0$. Let the point (h, k) be the image of the centroid of ΔABC in the line $3x + 6y - 53 = 0$. Then $h^2 + k^2 + hk$ is equal to:

(1) 40 (2) 36

(3) 47 (4) 37

Answer (4)

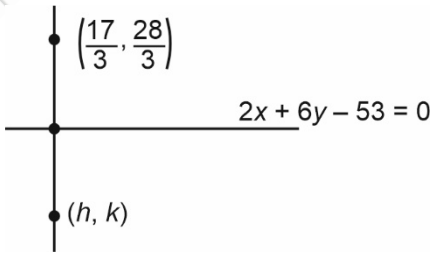
Sol.



Points of intersections are (9, 11), (3, 4), (5, 13)

Centroid of $\Delta ABC = \left(\frac{17}{3}, \frac{28}{3}\right)$

Since during image of ΔABC about line will reflect the whole triangle including centroid, reflected centroid will be image of $\left(\frac{17}{3}, \frac{28}{3}\right)$ about $2x + 6y - 53 = 0$



$$\frac{x - \frac{17}{3}}{2} = \frac{y - \frac{28}{3}}{6} = \frac{-2\left(2\left(\frac{17}{3}\right) + 6\left(\frac{28}{3}\right) - 53\right)}{2^2 + 6^2}$$

$\Rightarrow h = 3, k = 4$

$\Rightarrow h^2 + k^2 + hk = (h + k)^2 - hk$

$= 49 - 12 = 37$



3. Let $\vec{a}=2\hat{i}-\hat{j}+3\hat{k}$, $\vec{b}=3\hat{i}-5\hat{j}+\hat{k}$ and $\vec{c}$ be a vector such that $\vec{a}\times\vec{c}=\vec{c}\times\vec{b}$ and $(\vec{a}+\vec{c})\cdot(\vec{b}+\vec{c})=168$. Then the maximum value of $|\vec{c}|^2$ is

(1) 308 (2) 462

(3) 154 (4) 77

Answer (1)

Sol. $\vec{a}\times\vec{c}=\vec{c}\times\vec{b}=-(\vec{b}\times\vec{c})$

$\Rightarrow (\vec{a}\times\vec{c}+\vec{b}\times\vec{c})=(\vec{a}+\vec{b})\times\vec{c}=0$

$\Rightarrow$ $\vec{c}$ is parallel to $\vec{a}+\vec{b}$.

$\vec{c}=\lambda(5\hat{i}+(-6)\hat{j}+4\hat{k})\Rightarrow|\vec{c}|^2=\lambda^2(\pi)$

$(\vec{a}+\vec{c})\cdot(\vec{b}+\vec{c})=\vec{a}\cdot\vec{b}+|\vec{c}|^2+\vec{c}\cdot(\vec{a}+\vec{b})$

$=(6+5+3)+|\vec{c}|^2+\lambda(25+36+16)=168$

$\Rightarrow |\vec{c}|^2+\lambda(77)=154$

$\Rightarrow 77\lambda^2+77\lambda-152=0$

$\Rightarrow \lambda^2+\lambda-2=0$

$\Rightarrow (\lambda+2)(\lambda-1)=0$

$|\vec{c}|^2_{max}=(-2)^2(77)=308$

4. Let $y=y(x)$ be the solution of the differential equation $\cos x\left(\log_e(\cos x)\right)^2 dy$ $+(\sin x-3y\sin x\log_e(\cos x))dx=0$, $x\in\left(0,\frac{\pi}{2}\right)$. If $y\left(\frac{\pi}{4}\right)=\frac{-1}{\log_e 2}$, then $y\left(\frac{\pi}{6}\right)$ is equal to:

(1) $-\frac{1}{\log_e(4)}$ (2) $\frac{1}{\log_e(3)-\log_e(4)}$

(3) $\frac{1}{\log_e(4)-\log_e(3)}$ (4) $\frac{2}{\log_e(3)-\log_e(4)}$

Answer (2)

Sol. $\frac{dy}{dx}-\frac{3\sin x}{\cos x\ln\cos x}y=-\frac{\sin x}{\cos x(\ln\cos x)^2}$

I.F. = $e^{-\int\frac{3\tan x}{\ln\cos x}dx}$

Let ln cos$x$ = t

$-\tan x\,dx = dt$

$e^{3\int\frac{dt}{t}}=e^{3\ln t}=t^3=(\ln\cos x)^3$

$\therefore$ Solution will be

$y(\ln\cos x)^3=-\int(\tan x)(\ln\cos x)dx$

$y(\ln\cos x)^3=\frac{(\ln\cos x)^2}{2}+c$

$\because$ $y\left(\frac{\pi}{4}\right)=-\frac{1}{\ln 2}$

$\Rightarrow$ $c=0$

$\therefore$ $y=\frac{1}{2(\ln\cos x)}$

$y\left(\frac{\pi}{6}\right)=\frac{1}{2}\times\frac{1}{\ln\left(\cos\frac{\pi}{6}\right)}$

$=\frac{1}{2}\left[\frac{1}{\ln\left(\frac{\sqrt{3}}{2}\right)}\right]$

$=\frac{1}{2}\times\frac{1}{\ln\sqrt{3}-\ln 2}$

$=\frac{1}{\ln 3-\ln 4}$

5. The value of $\lim\limits_{n\to\infty}\left(\sum\limits_{k=1}^{n}\frac{k^3+6k^2+11k+5}{(k+3)!}\right)$ is

(1) $\frac{7}{3}$ (2) $\frac{4}{3}$

(3) $\frac{5}{3}$ (4) 2

Answer (3)



Sol. $\lim\limits_{n\to\infty}\left(\sum\limits_{k=1}^{n}\dfrac{k^3+6k^2+11k+5}{(k+3)!}\right)$

$\lim\limits_{n\to\infty}\left(\sum\limits_{k=1}^{n}\dfrac{k^3+6k^2+11k+6-1}{(k+3)!}\right)$

$\lim\limits_{n\to\infty}\left(\sum\limits_{k=1}^{n}\dfrac{(k+1)(k+2)(k+3)-1}{(k+3)!}\right)$

$\lim\limits_{n\to\infty}\left(\sum\limits_{k=1}^{n}\dfrac{1}{k!}-\dfrac{1}{(k+3)!}\right)$

$= \left(\dfrac{1}{1!}+\dfrac{1}{2!}+\dfrac{1}{3!}+\cdots\infty\right)-\left(\dfrac{1}{4!}+\dfrac{1}{5!}+\dfrac{1}{6!}+\cdots\infty\right)$

$= (e-1)-\left(e-1-\dfrac{1}{1!}-\dfrac{1}{2!}-\dfrac{1}{3!}\right)$

$= 1+\dfrac{1}{2}+\dfrac{1}{6}=\dfrac{10}{6}=\dfrac{5}{3}$

6. Let P be the set of seven digit numbers with sum of their digits equal to 11. If the numbers in P are formed by using the digits 1, 2 and 3 only, then the number of elements in the set P is:

(1) 173 (2) 161

(3) 164 (4) 158

Answer (2)

Sol. Case I: 3 2 2 1 1 1 1

$n_1=\dfrac{7!}{4!2!}=105$

Case II: 2 2 2 2 1 1 1

$n_2=\dfrac{7!}{4!3!}=35$

Case III: 3 3 1 1 1 1 1

$n_3=\dfrac{7!}{2!4!}=21$

Total numbers = 105 + 35 + 21 = 161

7. Let M and m respectively be the maximum and the minimum values of

$$f(x)=\begin{vmatrix}1+\sin^2 x & \cos^2 x & 4\sin 4x\\ \sin^2 x & 1+\cos^2 x & 4\sin 4x\\ \sin^2 x & \cos^2 x & 1+4\sin 4x\end{vmatrix},\ x\in R$$

Then $M^4 - m^4$ is equal to

(1) 1295 (2) 1040

(3) 1280 (4) 1215

Answer (3)

Sol. $R_1 \to R_1 - R_2,\ R_2 \to R_2 - R_3$

$$\begin{vmatrix}1 & -1 & 0\\ 0 & 1 & -1\\ \sin^2 x & \cos^2 x & 1+4\sin 4x\end{vmatrix}=f(x)$$

$C_2 \to C_1 + C_2$

$$\begin{vmatrix}1 & 0 & 0\\ 0 & 1 & -1\\ \sin^2 x & 1 & 1+4\sin 4x\end{vmatrix}=f(x)$$

$f(x) = 1 + 4\sin4x+1 = 2 + 4\sin4x$

$M = 6,\ m = -2$

$M^4 - m^4 = 1280$

8. The integral $80\int\limits_0^{\pi/4}\left(\dfrac{\sin\theta+\cos\theta}{9+16\sin 2\theta}\right)d\theta$ is equal to

(1) $4\log_e 3$ (2) $6\log_e 4$

(3) $2\log_e 3$ (4) $3\log_e 4$

Answer (1)

Sol. $I=\int\limits_0^{\frac{\pi}{4}}\left(\dfrac{\sin\theta+\cos\theta}{9-16\sin 2\theta}\right)d\theta$

Take $\sin\theta - \cos\theta = t$

$(\cos\theta + \sin\theta)d\theta = dt$

$(\sin\theta - \cos\theta)^2 = t^2$

$\Rightarrow \sin2\theta = 1 - t^2$

$\theta = 0 \to t = -1$

$\theta=\dfrac{\pi}{4}\to t=0$

$I=\int\limits_{-1}^{0}\dfrac{dt}{9+16(1-t^2)}$

$=\dfrac{1}{16}\int\limits_{-1}^{0}\dfrac{dt}{\frac{25}{16}-t^2}$

$=\dfrac{1}{4}\left[\dfrac{1}{10}\log\left|\dfrac{5+4t}{5-4t}\right|\right]_{-1}^{0}$

$=\dfrac{1}{40}[0+\log_e 9]$

$I=\dfrac{\log_e 9}{40}$

$80I = 2\log_e 9$

$80I = 4\log_e 3$

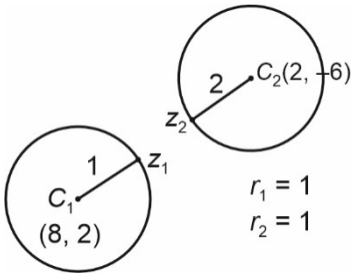


9. Let $|z_1 - 8 - 2i| \le 1$ and $|z_2 - 2 + 6i| \le 2$, $z_1, z_2 \in C$. Then the minimum value of $|z_1 - z_2|$ is

(1) 13 (2) 10
(3) 7 (4) 4

Answer (3)

Sol.



$r_1 = 1$
$r_2 = 1$

$|z_1 - z_2| = C_1C_2 - (r_1 + r_2)$

$= \sqrt{6^2 + 8^2} - (1 + 2)$

$= 10 - 3 = 7$

10. Let $A = [a_{ij}] = \begin{bmatrix} \log_5 128 & \log_4 5 \\ \log_5 8 & \log_4 25 \end{bmatrix}$.

If A_{ij} is the cofactor of $a_{ij}, C_{ij} = \sum_{k=1}^{2} a_{ik} A_{jk}, 1 \le i, j \le 2$ and $C = [C_{ij}]$, then $8|C|$ is equal to

(1) 222 (2) 242
(3) 288 (4) 262

Answer (2)

Sol. $[a_{ij}] = \begin{bmatrix} 7\log_5 2 & \frac{1}{2}\log_2 5 \\ 3\log_5 2 & \log_2 5 \end{bmatrix}$

$[A_{ij}] = \begin{bmatrix} \log_2 5 & -3\log_5 2 \\ -\frac{1}{2}\log_2 5 & 7\log_5 2 \end{bmatrix}$

$C_{ij} = \sum_{k=1}^{2} a_{ik} A_{jk}, \ 1 \le i, j \le 2$

$C = \begin{bmatrix} \frac{11}{2} & 0 \\ 0 & \frac{11}{2} \end{bmatrix}$

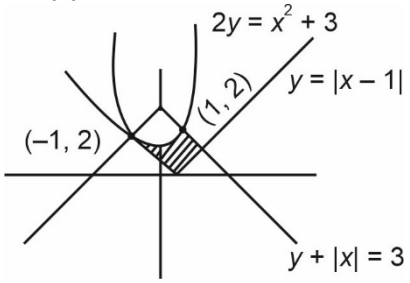
$8|C| = \frac{121}{4} \cdot 8 = 242$

11. Let the area of the region $\left\{(x, y) : 2y \le x^2 + 3, y + |x| \le 3, y \ge |x - 1|\right\}$ be A. Then $6A$ is equal to

(1) 18 (2) 12
(3) 16 (4) 14

Answer (4)

Sol.



$\int_{-1}^{1} \left(\frac{x^2 + 3}{2} + (x - 1)\right) dx + \int_{1}^{2} ((3 - x) - (x - 1)) dx$

$\Rightarrow \left. \frac{x^3}{6} + \frac{3x}{2} + \frac{x^2}{2} - x \right|_{-1}^{1} + \left. 3x - \frac{x^2}{2} - \frac{x^2}{2} + x \right|_{1}^{2}$

$A = \frac{7}{3} \Rightarrow 6A = 14$

12. Let $L_1 : \frac{x - 1}{1} = \frac{y - 2}{-1} = \frac{z - 1}{2}$ and $L_2 : \frac{x + 1}{-1} = \frac{y - 2}{2} = \frac{z}{1}$ be two lines.

Let L_3 be a line passing through the point (α, β, γ) and be perpendicular to both L_1 and L_2. If L_3 intersects L_1, then $|5\alpha - 11\beta - 8\gamma|$ equals

(1) 18 (2) 25
(3) 16 (4) 20

Answer (2)

Sol. $\frac{x - \alpha}{a} = \frac{y - \beta}{b} = \frac{z - \gamma}{c} : L_3$

$\left(a\hat{i} + b\hat{j} + c\hat{k}\right) \| \left(\hat{i} - \hat{j} + 2\hat{k}\right) \times \left(-\hat{i} + 2\hat{j} + \hat{k}\right)$

$(a, b, c) \equiv (5, 3, -1)$

$\Rightarrow \frac{x - \alpha}{5} = \frac{y - \beta}{3} = \frac{z - \gamma}{-1} = \lambda$

Point of intersection

$\alpha = p + 1 - 5\lambda, \beta = -p + 2 - 3\lambda$

$\gamma = 2p + 1 + \lambda$

$\Rightarrow |5\alpha - 11\beta - 8\gamma| = 25$



13. Let $x_1, x_2, \ldots, x_{10}$ be ten observations such that $\sum_{i=1}^{10}(x_i-2)=30$, $\sum_{i=1}^{10}(x_i-\beta)^2=98, \beta>2$, and their variance is $\frac{4}{5}$. If μ and σ^2 are respectively the mean and the variance of $2(x_1-1)+4\beta, 2(x_2-1)+4\beta, \ldots, 2(x_{10}-1)+4\beta$, then $\frac{\beta\mu}{\sigma^2}$ is equal to

(1) 110 (2) 90

(3) 100 (4) 120

Answer (3)

Sol. $\sum_{i=1}^{10}(x_i-2)=30$

$\sum_{i=1}^{10}x_i=50$

$\Rightarrow$ Mean = 5

Variance = $\frac{4}{5}=\frac{\sum x_i^2}{10}-\left(\bar{x}\right)^2$

$\frac{4}{5}=\frac{\sum x_i^2}{10}-25$

$\Rightarrow \sum x_i^2=258$

Now, $\sum_{i=1}^{10}(x_i-\beta)^2=98$

$\sum_{i=1}^{10}x_i^2-2\beta\sum_{i=1}^{10}x_i+10\beta^2=98$

$\Rightarrow$ $258 - 2\beta(50) + 10\beta^2 = 98$

$\Rightarrow$ $10\beta^2 - 100\beta + 160 = 0$

$\Rightarrow$ $\beta^2 - 10\beta + 16 = 0$

$\Rightarrow$ $\beta = 8$ as $\beta > 2$

Now, as per the question

$2(x_1 - 1) + 4\beta, 2(x_2 - 1) + 4\beta, \ldots 2(x_{10} - 1) + 4\beta$

Can be simplified as

$2x_1 + 30, 2x_2 + 30, \ldots, 2x_{10} + 30$

$\mu = 2(5) + 30 = 40$

$\sigma^2 = 2^2\left(\frac{4}{5}\right)=\frac{16}{5}$

$\frac{\beta\mu}{\sigma^2}=\frac{8\times40}{\frac{16}{5}}=100$

14. The least value of n for which the number of integral terms in the Binomial expansion of $\left(\sqrt[3]{7}+\sqrt[12]{11}\right)^n$ is 183, is

(1) 2184 (2) 2148

(3) 2196 (4) 2172

Answer (1)

Sol. $\left(\sqrt[3]{7}+\sqrt[12]{11}\right)^n$

$T_{K+1}={}^nC_K 7^{\frac{n-k}{3}}11^{\frac{k}{12}}$

12 divides k and 3 divides $n - k$

For integer terms

$\Rightarrow$ Multiples of 12 for k would work

$\Rightarrow$ $k = 0, 12, 24$

$\Rightarrow$ $k_{max} = 12 \times 182 = 2184$

$\Rightarrow$ Minimum value of n will be 2184

$\Rightarrow$ Option (1) is correct

15. The number of solutions of the equation $\left(\frac{9}{x}-\frac{9}{\sqrt{x}}+2\right)\left(\frac{2}{x}-\frac{7}{\sqrt{x}}+3\right)=0$ is

(1) 4 (2) 3

(3) 1 (4) 2

Answer (1)

Sol. $\left(\frac{9}{x}-\frac{9}{\sqrt{x}}+2\right)\left(\frac{2}{x}-\frac{7}{\sqrt{x}}+3\right)=0$

Let $\frac{1}{\sqrt{x}}=t$

$(9t^2 - 9t + 2)(2t^2 - 7t + 3) = 0$



$9t^2 - 9t + 2 = 0$

$\Rightarrow t = \frac{9 \pm \sqrt{81-72}}{18}$

$\Rightarrow t = \frac{9 \pm 3}{18}$

$\Rightarrow t = \frac{2}{3}, \frac{1}{3}$

$\Rightarrow 2t^2 - 7t + 3 = 0$

$\Rightarrow 2t^2 - 6t - t + 3 = 0$

$\Rightarrow 2t(t-3) - 1(t-3) = 0$

$t = \frac{1}{2}, 3$

$\Rightarrow \sqrt{x} = \frac{3}{2}, 3, 2, \frac{1}{3}$

$\Rightarrow x = \frac{9}{4}, 9, 4, \frac{1}{9}$

Hence, number of solutions = 4

16. Consider an A.P. of positive integers, whose sum of the first three terms is 54 and the sum of the first twenty terms lies between 1600 and 1800. Then its 11th term is:

(1) 84 (2) 122

(3) 90 (4) 108

Answer (3)

Sol. Let A.P. be a, $a + d$, $a + 2d$

$3a + 3d = 54$

$a + d = 18$ ... (1)

$1600 < \frac{20}{2}[2a + 19d] < 1800$

$160 < 2a + 19d < 180$

$160 < 18 \times 2 + 17d < 180$

$\frac{124}{7} < d < \frac{144}{17}$

$\because d \in$ integer $\Rightarrow d = 8$

$a + d = 18$

$\Rightarrow a = 10$

$a_{11} = a + 10d$

$= 10 + 10 \times 8 = 90$

17. Let the ellipse $E_1 : \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, $a > b$ and $E_2 : \frac{x^2}{A^2} + \frac{y^2}{B^2} = 1$, $A < B$ have same eccentricity $\frac{1}{\sqrt{3}}$. Let the product of their lengths of latus rectums be $\frac{32}{\sqrt{3}}$, and the distance between the foci of E_1 be 4. If E_1 and E_2 meet at A, B, C and D, then the area of the quadrilateral $ABCD$ equals

(1) $6\sqrt{6}$ (2) $\frac{24\sqrt{6}}{5}$

(3) $\frac{12\sqrt{6}}{5}$ (4) $\frac{18\sqrt{6}}{5}$

Answer (2)

Sol. $2ae = 4$

$\Rightarrow a = 2\sqrt{3}$

$\Rightarrow 1 - \frac{b^2}{12} = \frac{1}{3} \Rightarrow b^2 = 8$

$\frac{2b^2}{a} \times \frac{2A^2}{B} = \frac{32}{\sqrt{3}}$

$\Rightarrow \frac{2 \times 8}{2\sqrt{3}} \times \frac{2A^2}{B} = \frac{32}{\sqrt{3}}$

$\Rightarrow \frac{A^2}{B} = 2 \Rightarrow A^2 = 2B$

$1 - \frac{A^2}{B} = \frac{1}{3}$

$\Rightarrow B = 3 \Rightarrow A^2 = 6$

$E_1 : \frac{x^2}{12} + \frac{y^2}{8} = 1$ ...(i)

$E_1 : \frac{x^2}{6} + \frac{y^2}{9} = 1$ ...(ii)

On solving (i) & (ii)

$(x, y) = \left(\frac{\sqrt{6}}{\sqrt{5}}, \frac{6}{\sqrt{5}}\right), \left(\frac{-\sqrt{6}}{\sqrt{5}}, \frac{6}{\sqrt{5}}\right), \left(\frac{\sqrt{6}}{\sqrt{5}}, \frac{-6}{\sqrt{5}}\right), \left(\frac{-\sqrt{6}}{\sqrt{5}}, \frac{-6}{\sqrt{5}}\right)$

Four points are vertices of rectangle area = $\frac{24\sqrt{6}}{5}$



18. Let $\vec{a} = \hat{i} + 2\hat{j} + \hat{k}$ and $\vec{b} = 2\hat{i} + 7\hat{j} + 3\hat{k}$. Let $L_1 : \vec{r} = \left(-\hat{i} + 2\hat{j} + \hat{k}\right) + \lambda\vec{a}, \lambda \in \mathbf{R}$ and $L_2 : \vec{r} = \left(\hat{j} + \hat{k}\right) + \mu\vec{b}, \mu \in \mathbf{R}$ be two lines. If the line L_3 passes through the point of intersection of L_1 and L_2 and is parallel to $\vec{a} + \vec{b}$, then L_3 passes through the point

(1) (8, 26, 12) (2) (2, 8, 5)

(3) (5, 17, 4) (4) (–1, –1, 1)

Answer (1)

Sol. $\vec{a} = \hat{i} + 2\hat{j} + \hat{k}$

$\vec{b} = 2\hat{i} + 7\hat{j} + 3\hat{k}$

$L_1 : \vec{r} = \left(-\hat{i} + 2\hat{j} + \hat{k}\right) + \lambda\vec{a}, \lambda \in R$

and $L_2 = \left(\hat{j} + \hat{k}\right) + \mu\vec{b}, \mu \in R$

$\vec{a} + \vec{b} = 3\hat{i} + 9\hat{j} + 4\hat{k}$

$L_1 : \vec{r} = (\lambda - 1)\hat{i} + 2(\lambda + 1)\hat{j} + (\lambda + 1)\hat{k}$

$L_2 : \vec{r} = 2\mu\hat{i} + (7\mu + 1)\hat{j} + (1 + 3\mu)\hat{k}$

For point of intersection of L_1 & L_2

$\lambda - 1 = 2\mu$ and $2(\lambda + 1) = 7\mu + 1$

$\Rightarrow \lambda = 3$ and $\mu = 1$

$L_3 : \vec{r} = 2\hat{i} + 8\hat{i} + 4\hat{k} + \alpha(3\hat{i} + 9\hat{j} + 4\hat{k})$

For $\alpha = 2$

$\vec{r} = 8\hat{i} + 26\hat{j} + 12\hat{k}$

19. Two parabolas have the same focus (4, 3) and their directrices are the x-axis and the y-axis, respectively. If these parabolas intersects at the points A and B, then $(AB)^2$ is equal to :

(1) 392 (2) 192

(3) 384 (4) 96

Answer (2)

Sol. The parabolas are

$(x - 4)^2 + (y - 3)^2 = x^2$ …(i)

and $(x - 4)^2 + (y - 3)^2 = y^2$ …(ii)

If point of intersection are $A(x_1, y_1)$ and $B(x_2, y_2)$

By solving (i) and (ii), we get

$x_1 + x_2 = 14$ and $x_1 x_2 = 25$

$(AB)^2 = 2((x_1 + x_2)^2 - 4 x_1 x_2) = 192$

20. Let the line $x + y = 1$ meet the circle $x^2 + y^2 = 4$ at the points A and B. If the line perpendicular to AB and passing through the mid-point of the chord AB intersects the circle at C and D, then the area of the quadrilateral $ADBC$ is equal to:

(1) $2\sqrt{14}$

(2) $3\sqrt{7}$

(3) $5\sqrt{7}$

(4) $\sqrt{14}$

Answer (1)

Sol. Solving $x = y$ & $x^2 + y^2 = 4$ gives

$C\left(\sqrt{2}, \sqrt{2}\right)$ and $D\left(-\sqrt{2}, -\sqrt{2}\right)$

Solving $x + y = 1$ & $x^2 + y^2 = 4$ gives

$A\left(\frac{1+\sqrt{7}}{2}, \frac{1-\sqrt{7}}{2}\right)$ & $B\left(\frac{1-\sqrt{7}}{2}, \frac{1+\sqrt{7}}{2}\right)$

Required area $= 2 \times \frac{1}{2}\begin{vmatrix} \sqrt{2} & \sqrt{2} & 1 \\ \frac{1-\sqrt{7}}{2} & \frac{1+\sqrt{7}}{2} & 1 \\ -\sqrt{2} & -\sqrt{2} & 1 \end{vmatrix}$

$= 2\sqrt{14}$ sq. units



SECTION - B

Numerical Value Type Questions: This section contains 5 Numerical based questions. The answer to each question should be rounded-off to the nearest integer.

21. Let $[t]$ be the greatest integer less than or equal to t. Then the least value of $p \in N$ for which

$$\lim_{x\to 0^+}\left(x\left(\left[\frac{1}{x}\right]+\left[\frac{2}{x}\right]+\ldots+\left[\frac{p}{x}\right]\right)-x^2\left(\left[\frac{1}{x^2}\right]+\left[\frac{2^2}{x^2}\right]+\ldots\left[\frac{9^2}{x^2}\right]\right)\right)\geq 1$$

is equal to _____.

Answer (24)

Sol. $\lim_{x\to 0^+}\left(x\left[\frac{1}{x}\right]+\left[\frac{2}{x}\right]+\ldots+\left[\frac{p}{x}\right]\right)-x^2\left(\left[\frac{1}{x^2}\right]+\left[\frac{2^2}{x^2}\right]+\ldots+\left[\frac{9^2}{x^2}\right]\right)\geq 1$

$\Rightarrow (1 + 2 + 3 +\ldots + p) - (1^2 + 2^2 + \ldots + 9^2) \geq 1$

$\Rightarrow \frac{p(p+1)}{2}-\frac{9(10)(19)}{6}\geq 1$

$\Rightarrow p(p + 1) \geq 572$

$\Rightarrow$ Least natural values of p is 24

22. Let $f:(0,\infty)\to \mathbf{R}$ be a twice differentiable function.

If for some $a \neq 0, \int_0^1 f(\lambda x)d\lambda = af(x), f(1) = 1$ and $f(16)=\frac{1}{8}$, then $16-f'\left(\frac{1}{16}\right)$ is equal to____.

Answer (112)

Sol. Given, $\int_0^1 f(\lambda x)d\lambda = af(x)$ ...(1)

Let $\lambda x = u$

$d\lambda = \frac{1}{x}du$

$\therefore$ From (1) $\frac{1}{x}\int_0^x f(u)du = af(x)$

$\Rightarrow \int_0^x f(u)du = axf(x)$

Differentiate both sides

$f(x) = a(xf'(x) + f(x))$

$\Rightarrow f(x) = axf'(x) + af(x)$

$\Rightarrow (1-a)f(x) = axf'(x)$

$\Rightarrow \frac{f'(x)}{f(x)}=\frac{(1-a)}{a}\cdot\frac{1}{x}$

Integrate both side w.r.t. (x)

$\Rightarrow \int\frac{f'(x)}{f(x)}dx=\frac{(1-a)}{a}\int\frac{1}{x}dx$

$\Rightarrow \ln f(x)=\left(\frac{1-a}{a}\right)\ln x + c$

Now at $x = 1$ $f(1) = 1$

$\Rightarrow c = 0$

Also given $f(16)=\frac{1}{8}$

$\therefore \frac{1}{8}=(16)^{\frac{1-a}{a}}$

$\Rightarrow 2^{-3}=2^{\frac{4-4a}{a}}$

$\Rightarrow -3=\frac{4-4a}{a}$

$\Rightarrow -3a = 4 - 4a$

$\Rightarrow a = 4$

$\therefore f(x) = x^{-3/4}$

$f(x)=\frac{-3}{4}x^{-\frac{7}{4}}$

Put $x=\frac{1}{16}$

$f'\left(\frac{1}{16}\right)=\frac{-3}{4}\left(\frac{1}{16}\right)^{-7/4}=\frac{-3}{4}\cdot 2^{-4\times\left(\frac{-7}{4}\right)}=-96$

$\therefore 16-f'\left(\frac{1}{16}\right)\Rightarrow 16-(-96)=112$



23. Let $S = \{x : \cos^{-1} x = \pi + \sin^{-1} x + \sin^{-1}(2x+1)\}$.

Then $\sum_{x \in S} (2x-1)^2$ is equal to_____.

Answer (5)

Sol. $\cos^{-1} x = \pi + \sin^{-1} x + \sin^{-1}(2x+1)$.

$$\frac{\pi}{2} - \sin^{-1} x = \pi + \sin^{-1} x + \sin^{-1}(2x+1)$$

$$-\frac{\pi}{2} - 2\sin^{-1} x = \sin^{-1}(2x+1)$$

$$\sin\left(-\frac{\pi}{2} - 2\sin^{-1} x\right) = \sin\left(\sin^{-1}(2x+1)\right)$$

$-\cos(2\sin^{-1}x) = (2x + 1)$

$-(1 - 2x^2) = 2x + 1$

$-1 + 2x^2 = 2x + 1$

$2x^2 - 2x - 2 = 0$

$$x = \frac{2 \pm \sqrt{4+16}}{4}$$

$$x = \frac{2 \pm 2\sqrt{5}}{4} = \frac{1 \pm \sqrt{5}}{2} \left\{x = \frac{1+\sqrt{5}}{2} \text{ rejected}\right\}$$

So, $\sum_{x \in s} (2x-1)^2 = 5$

24. The number of 6-letter words, with or without meaning, that can be formed using the letters of the word MATHS such that any letter that appears in the word must appear at least twice is

Answer (1405)

Sol. $${}^5C_3 \times \frac{6!}{2!2!2!} + {}^5C_2\left(\frac{6!}{2!4!} \times 2 + \frac{6!}{3!3!}\right) + {}^5C_1 \cdot 1$$

$= 10 \times 90 + 10(15 \times 2 + 20) + 5$

$= 900 + 500 + 5$

$= 1405$

25. Let $S = \left\{m \in \mathbf{Z} : A^{m^2} + A^m = 3I - A^{-6}\right\}$, where $A = \begin{bmatrix} 2 & -1 \\ 1 & 0 \end{bmatrix}$. Then $n(S)$ is equal to_____.

Answer (2)

Sol. $A = \begin{bmatrix} 2 & -1 \\ 1 & 0 \end{bmatrix}$

Now finding characteristic equation

$$\begin{vmatrix} 2-\lambda & -1 \\ 1 & -\lambda \end{vmatrix} = 0$$

$\Rightarrow (2 - \lambda)(-\lambda) - (-1)(1) = -2\lambda + \lambda^2 + 1 = 0$

$\Rightarrow \lambda^2 - 2\lambda + 1 = 0$

$\Rightarrow (\lambda - 1)^2 = 0$

$\Rightarrow \lambda = 1$

Since A satisfies $(A - I)^2 = 0$

$\therefore$ $A = I + N$ where

$N = A - I$

$$N = \begin{bmatrix} 1 & -1 \\ 1 & -1 \end{bmatrix}$$

$N^2 = 0$

$A^m = (I + N)^m = I + mN$

$A^m \cdot A^m = (I + mN)(I + mN) = I + 2mN + m^2N^2$

Since $N^2 = 0$

$\Rightarrow A^{m^2} = I + 2mN$

Now putting in given condition

$I + m^2N + I + mN = 3I - A^{-6}$ ...(i)

$$A^{-1} = \begin{bmatrix} 0 & 1 \\ -1 & 2 \end{bmatrix}$$

$A^{-6} = (A^{-1})^6 = I + (-6)N$

$\therefore$ Putting in (i)

$(m^2 + m)N = I - (I - 6N)$

$(m^2 + m)N = 6N$

Since $N \neq 0$

$\Rightarrow m^2 + m = 6$

$\Rightarrow m^2 + m - 6 = 0$

$\Rightarrow (m - 2)(m + 3) = 0$

$\Rightarrow m = 2, -3$

$\therefore$ Number of elements in S is 2



PHYSICS

SECTION - A

Multiple Choice Questions: This section contains 20 multiple choice questions. Each question has 4 choices (1), (2), (3) and (4), out of which **ONLY ONE** is correct.

Choose the correct answer :

26. The workdone in an adiabatic change in an ideal gas depends upon only :
    (1) Change in its volume
    (2) Change in its temperature
    (3) Change in its pressure
    (4) Change in its specific heat

Answer (2)

Sol. Work done in adiabatic process = $\frac{nR\Delta T}{1-\gamma}$

So, depends upon change in temperature.

27. Two projectiles are fired with same initial speed from same point on ground at angles of $(45° - \alpha)$ and $(45° + \alpha)$, respectively, with the horizontal direction. The ratio of their maximum heights attained is :
    (1) $\frac{1+\sin 2\alpha}{1-\sin 2\alpha}$ (2) $\frac{1+\sin\alpha}{1-\sin\alpha}$
    (3) $\frac{1-\tan\alpha}{1+\tan\alpha}$ (4) $\frac{1-\sin 2\alpha}{1+\sin 2\alpha}$

Answer (4)

Sol. $H_{max} = \frac{u^2\sin^2\theta}{2g}$

$$\frac{H_1}{H_2} = \frac{\sin^2(45-\alpha)}{\sin^2(45+\alpha)}$$

$$= \frac{(\cos\alpha-\sin\alpha)^2}{(\cos\alpha+\sin\alpha)^2}$$

$$\frac{H_1}{H_2} = \frac{1-\sin 2\alpha}{1+\sin 2\alpha}$$

28. Consider a long straight wire of a circular cross-section (radius *a*) carrying a steady current *I*. The current is uniformly distributed across this cross-section. The distances from the centre of the wire's cross-section at which the magnetic field [inside the wire, outside the wire] is half of the maximum possible magnetic field, anywhere due to the wire, will be
    (1) [a/2, 2a] (2) [a/2, 3a]
    (3) [a/4, 2a] (4) [a/4, 3a/2]

Answer (1)

Sol. Magnetic field inside cylinder at distance *r* from axis

$$= \frac{\mu_0 I}{\pi R^2}\cdot\frac{r}{2}$$

$$B_{max} = \frac{\mu_0 I}{2\pi R}$$

$$\frac{\mu_0 I}{2\pi R^2} r_1 = \frac{1}{2}\left(\frac{\mu_0 I}{2\pi R}\right)$$

$$r_1 = \frac{R}{2} = \frac{a}{2}$$

Magnetic field outside cylinder at distance *r* from axis

$$= \frac{\mu_0 I}{2\pi r}$$

$$\frac{\mu_0 I}{2\pi r_2} = \frac{1}{2}\left(\frac{\mu_0 I}{2\pi R}\right)$$

$r_2 = 2R = 2a$

29. At the interface between two materials having refractive indices n_1 and n_2, the critical angle for reflection of an em wave is θ_{1C}. The n_2 material is replaced by another material having refractive index n_3 such that the critical angle at the interface between n_1 and n_3 materials is θ_{2C}. If $n_3 > n_2 > n_1$; $\frac{n_2}{n_3} = \frac{2}{5}$ and $\sin\theta_{2C} - \sin\theta_{1C} = \frac{1}{2}$, then θ_{1C} is
    (1) $\sin^{-1}\left(\frac{1}{3n_1}\right)$ (2) $\sin^{-1}\left(\frac{1}{6n_1}\right)$
    (3) $\sin^{-1}\left(\frac{5}{6n_1}\right)$ (4) $\sin^{-1}\left(\frac{2}{3n_1}\right)$

Answer (*)

